05 – C, gcc, libraries, and make

How to compile and run a C program?

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OUTLINE

1) C language
2) gcc
3) Libraries
    1) dynamic linking (default)
    2) static linking
4) Creating Libraries
5) Make

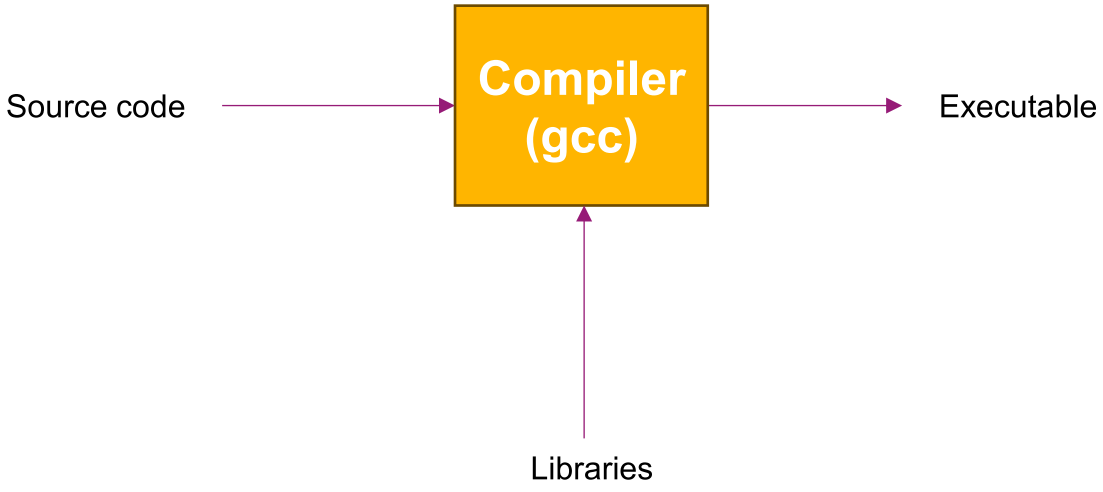
1) C LANGUAGE

- The C language has been invented by Dennis RITCHIE in 1972
- It is the language used to code the kernels of UNIX operating systems
- It is a low-level language: pointers, malloc(), free(), …

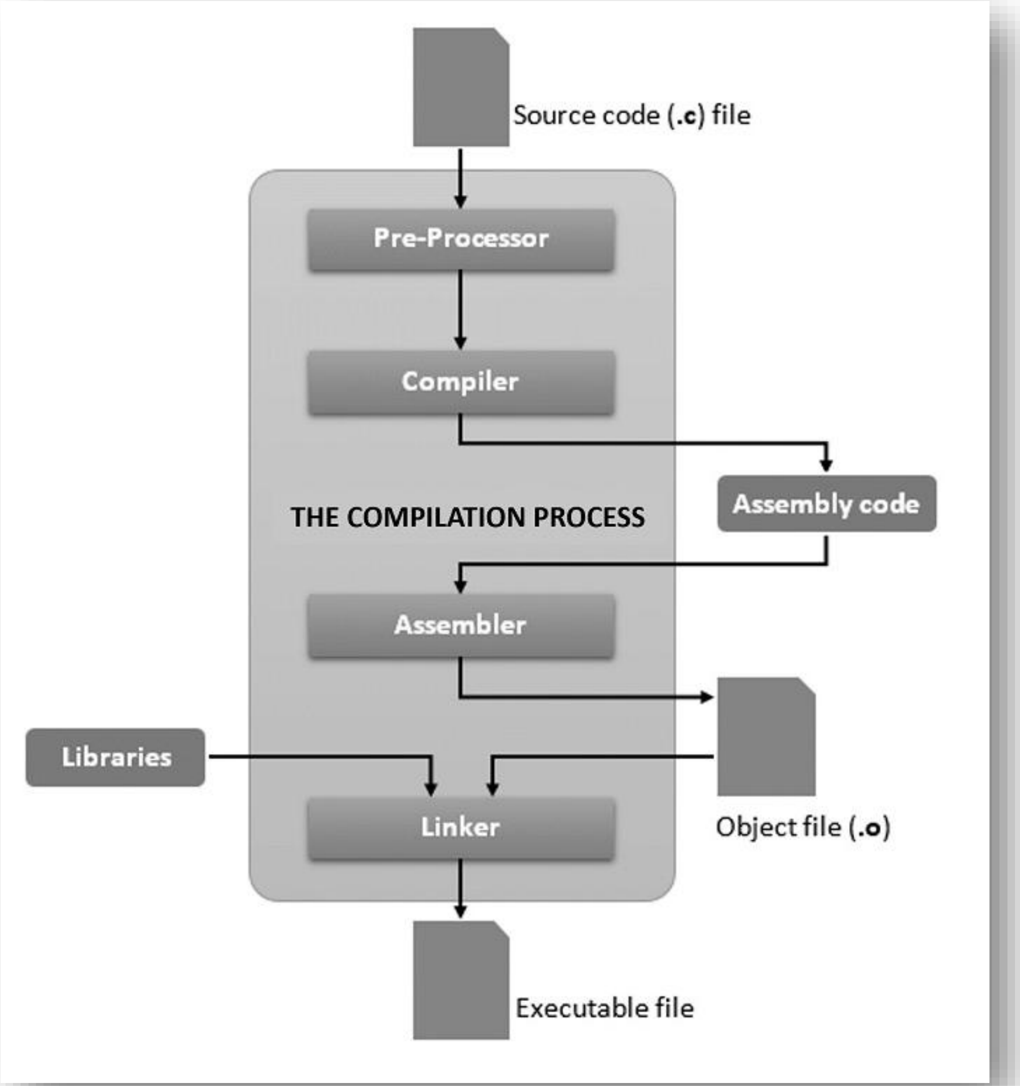
Source: https://en.wikipedia.org/wiki/C_(programming_language)

1) C LANGUAGE

- The C language is compiled and linked to libraries to build executable files



1) C LANGUAGE



Source: https://www.tutorialspoint.com/cprogramming/index.htm

2) GCC

- gcc is (Gnu Compiler Collection, it had GNU C Compiler for first name) is the compiler available on Linux
- It has been developed by the GNU (GNU is Not Unix) foundation
- It is a free and very powerful open-source compiler

2) GCC

- Let’s write a very ”simple” Hello World ”hello.c” file:

```c
#include <stdio.h>

int main(void) {
		printf("Hello, World!\n");
		return 0;
}
```

2) GCC

- To compile a ".c" file into a executable "a.out":

```
[jlandre@jlandre-LINUX c]$ vim hello.c
[jlandre@jlandre-LINUX c]$ ls -l
total 4
-rw-r--r-- 1 jlandre jlandre 79  6 nov 15.29 hello.c
[jlandre@jlandre-LINUX c]$ gcc hello.c
[jlandre@jlandre-LINUX c]$ ls -l
total 20
-rwxr-xr-x 1 jlandre jlandre 13840  6 nov 15.30 a.out
-rw-r--r-- 1 jlandre jlandre    79  6 nov 15.29 hello.c
[jlandre@jlandre-LINUX c]$ ./a.out
Hello, World!
[jlandre@jlandre-LINUX c]$ 
```

2) GCC

- To compile a ”.c” file into a executable ”hello”:

```
[jlandre@jlandre-LINUX c]$ gcc -o hello hello.c
[jlandre@jlandre-LINUX c]$ ls -l
total 36
-rwxr-xr-x 1 jlandre jlandre 13840  6 nov 15.30 a.out
-rwxr-xr-x 1 jlandre jlandre 13840  6 nov 15.32 hello
-rw-r--r-- 1 jlandre jlandre    79  6 nov 15.29 hello.c
[jlandre@jlandre-LINUX c]$ ./hello
Hello, World!
[jlandre@jlandre-LINUX c]$ 
```

2) GCC

- Creation of a library of function to use in a project, ”lib.c”:

```c
#include <stdio.h>
#include <time.h>
#include "lib.h"

void printDateTime(void) {
        time_t now;
        (void) time(&now);
        printf("%s", ctime(&now));
}

```

2) GCC

- Creation of a library of function to use in a project, header file ”lib.h”:

```c
// file lib.h

// contain the prototypes of the functions
void printDateTime(void);

```

2) GCC

- Use of the library in the main function:

```c
#include <stdio.h>
#include "lib.h"

int main(void) {
		printf("Hello, World!\n");
		printDateTime();
		return 0;
}

```

2) GCC

- Use of gcc to compile the file:

```
[jlandre@jlandre-LINUX c]$ gcc hello2.c lib.c -o hello2
[jlandre@jlandre-LINUX c]$ ./hello2
Hello, World!
Mon Nov  6 15:53:32 2023
[jlandre@jlandre-LINUX c]$ 
```

2) GCC

- Others options of gcc:

  - To add debug information: -g
  - To add warnings: -Wall

```
gcc -g -Wall -o program file1.c file2.c
```

2) GCC

- The linker (ld) can be used to compile object files one per one:

```
gcc -g -Wall -c module1.c
gcc -g -Wall -c module2.c
gcc -g -Wall -c main.c
gcc -o program main.o module1.o module2.o
```

Source: Reviewing gcc, make, gdb, and Linux Editors, Colin Gordon

2) GCC

- -o file Writes result to file
- -c Stops compilation with an object file; no linking
- -g Outputs debugging information
- -On Uses optimization level n, for 0 <= n <= 3

Uppercase i
as in include

- -I dir Looks for header files in dir - an include directory
- -L dir Looks for libraries to link against in dir

Lowercase l
as in lib

- -l lib Link against the library lib
- -Wall Warn about anything questionable
- -Werror Treat all warnings as compilation errors

3) LINKING

- When we use a library, we can choose between:
  - Dynamic linking (default)
  - Static linking

3) LINKING: DYNAMIC LINKING (DEFAULT)

- When you compile with gcc, by default, the linking of libraries is dynamic:

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ ls -l
total 4
-rw-rw-r-- 1 os2024 os2024 76 okt 22 15:28 hello-world.c
os2024@os2024-virtualbox:~/Desktop/c/linking$
```

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ gcc -o hello hello-world.c
os2024@os2024-virtualbox:~/Desktop/c/linking$ ls -l
total 20
-rwxrwxr-x 1 os2024 os2024 15960 okt 22 15:30 hello
-rw-rw-r-- 1 os2024 os2024    76 okt 22 15:28 hello-world.c
os2024@os2024-virtualbox:~/Desktop/c/linking$ ./hello
Hello, World!
os2024@os2024-virtualbox:~/Desktop/c/linking$
```

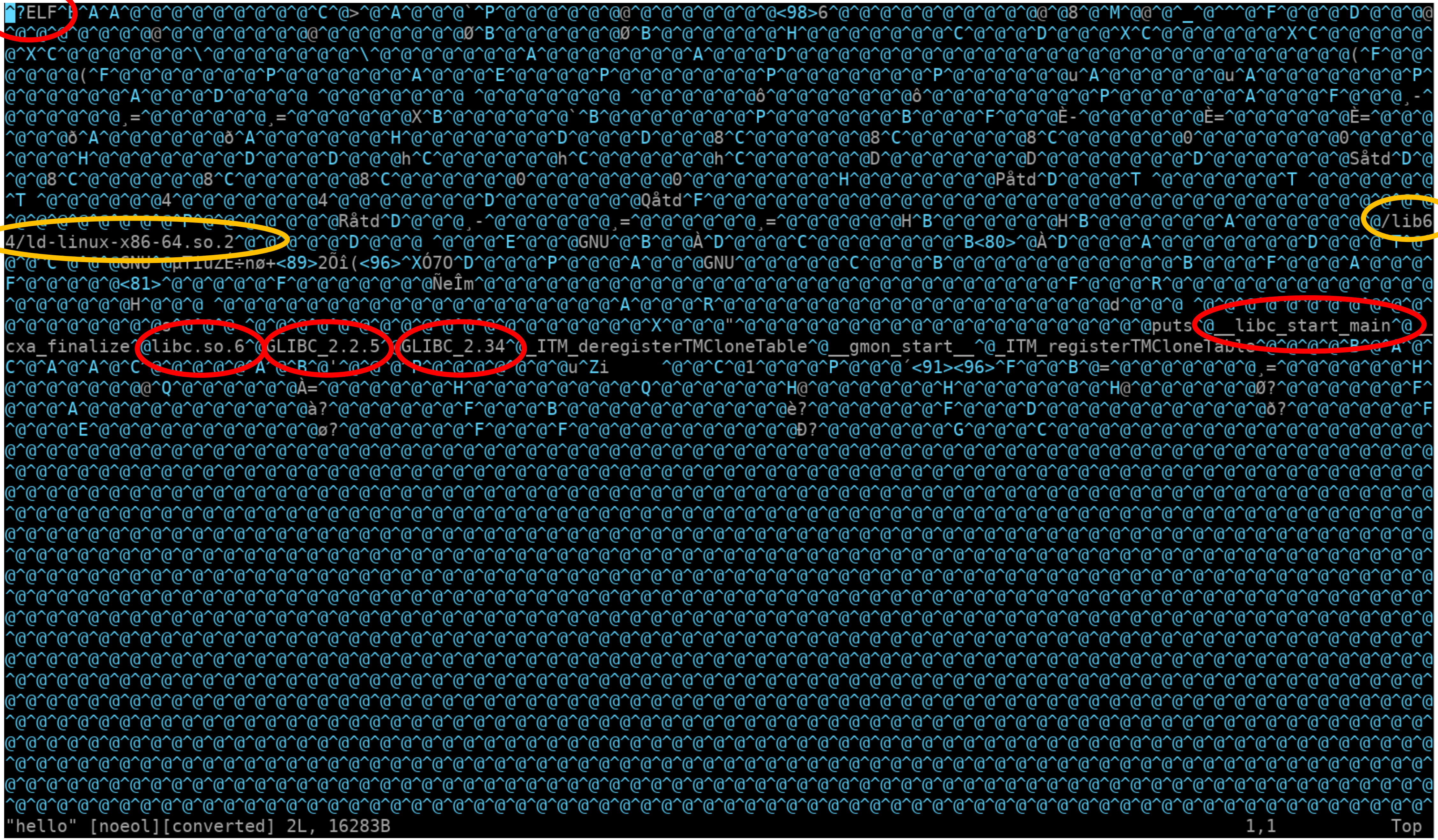
```
os2024@os2024-virtualbox:~/Desktop/c/linking$ vi hello
```

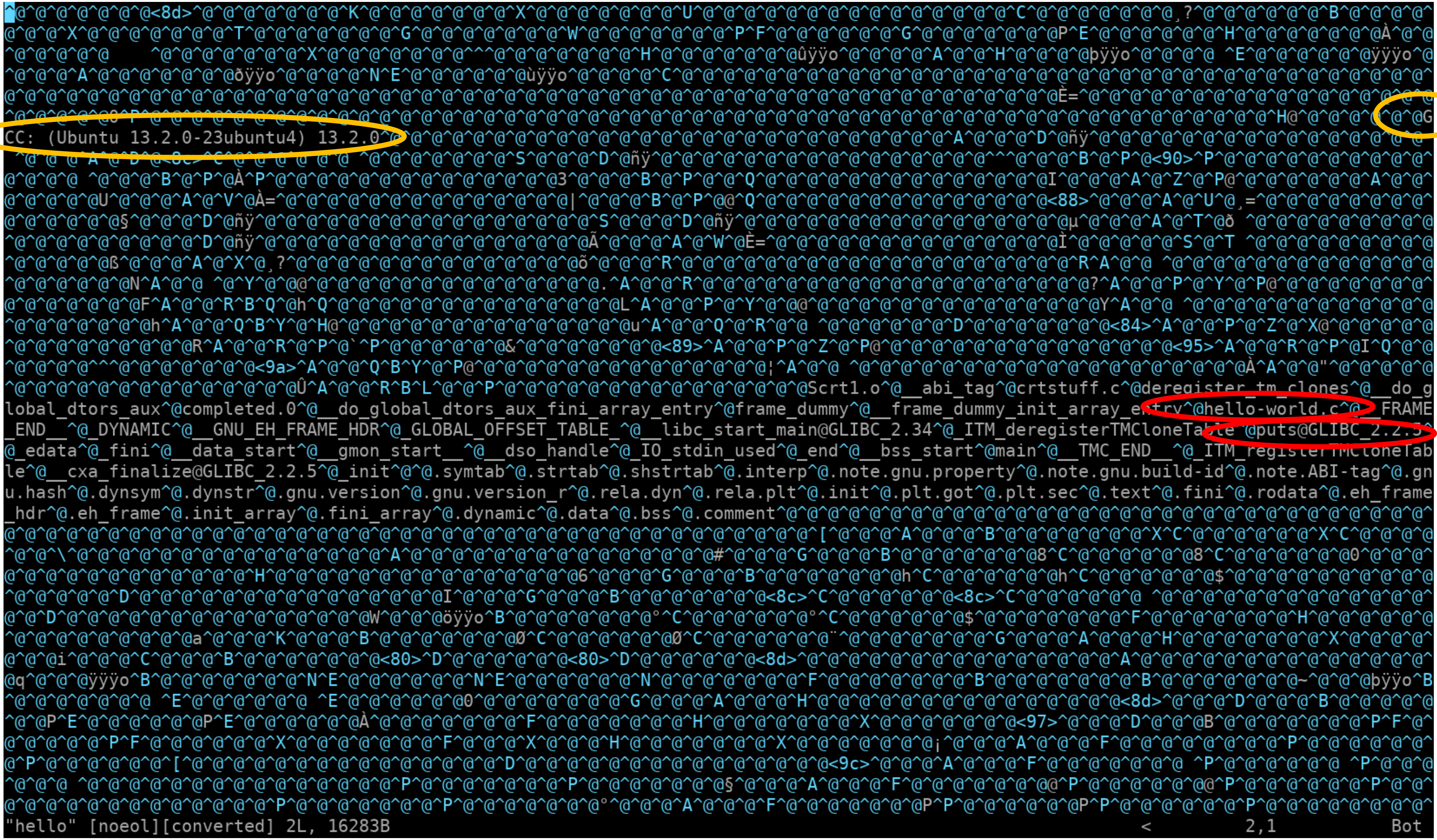
3) LINKING: DYNAMIC LINKING (DEFAULT)

- When you open the executable file with a text editor:

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ vi hello
```

- The resulting file is not ”human readable” **HOWEVER** there are some symbols that are readable and give interesting information on the executable format, content and needed libraries


^?ELF
/lib64/ld-linux-x86-64.so.2
GNU
puts
__libc_start_main
__cxa_finalize
libc.so.6
GLIBC_2.2.5
GLIBC_2.34
_ITM_deregisterTMCloneTable
__gmon_start__
_ITM_registerTMCloneTable
"hello" [noeol][converted] 2L, 16283B
1,1
Top



GCC: (Ubuntu 13.2.0-23ubuntu4) 13.2.0
Scrt1.o^@__abi_tag^@crtstuff.c^@deregister_tm_clones^@__do_global_dtors_aux^@completed.0^@__do_global_dtors_aux_fini_array_entry^@frame_dummy^@__frame_dummy_init_array_entry^@hello-world.c^@__FRAME_END__^@_DYNAMIC^@__GNU_EH_FRAME_HDR^@_GLOBAL_OFFSET_TABLE_^@__libc_start_main@GLIBC_2.34^@_ITM_deregisterTMCloneTable^@puts@GLIBC_2.2.5^@_edata^@_fini^@__data_start^@__gmon_start__^@__dso_handle^@_IO_stdin_used^@_end^@__bss_start^@main^@__TMC_END__^@_ITM_registerTMCloneTable^@__cxa_finalize@GLIBC_2.2.5^@_init^@^@.symtab^@.strtab^@.shstrtab^@.interp^@.note.gnu.property^@.note.gnu.build-id^@.note.ABI-tag^@.gnu.hash^@.dynsym^@.dynstr^@.gnu.version^@.gnu.version_r^@.rela.dyn^@.rela.plt^@.init^@.plt.got^@.plt.sec^@.text^@.fini^@.rodata^@.eh_frame_hdr^@.eh_frame^@.init_array^@.fini_array^@.dynamic^@.data^@.bss^@.comment
"hello" [noeol][converted] 2L, 16283B
2,1
Bot

3) LINKING: STATIC LINKING

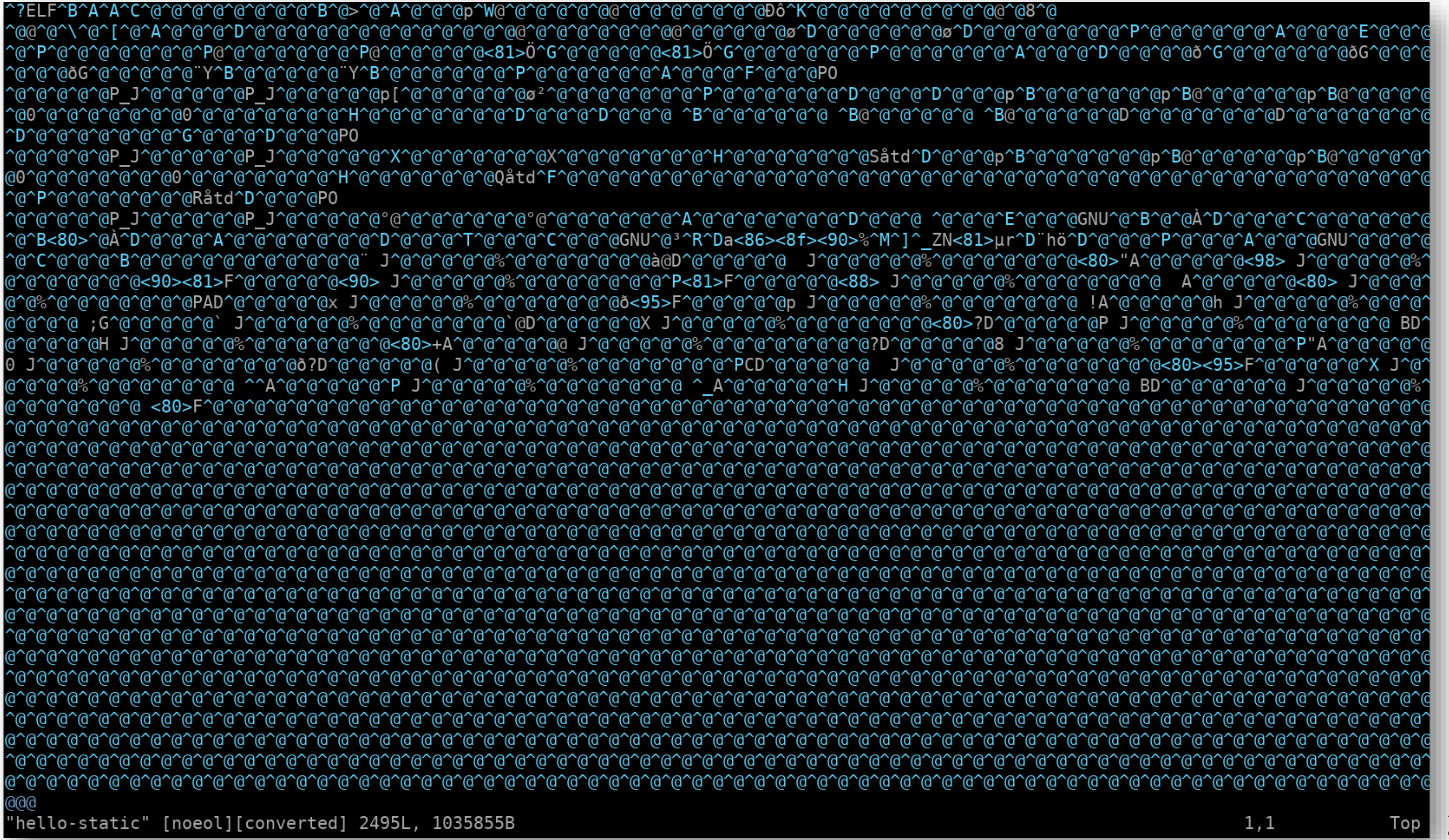
```
os2024@os2024-virtualbox:~/Desktop/c/linking$ gcc -static -o hello-static hello-world.c
os2024@os2024-virtualbox:~/Desktop/c/linking$ ls -l
total 788
-rwxrwxr-x 1 os2024 os2024  15960 okt 22 15:30 hello
-rwxrwxr-x 1 os2024 os2024 785360 okt 22 15:49 hello-static
-rw-rw-r-- 1 os2024 os2024     76 okt 22 15:28 hello-world.c
os2024@os2024-virtualbox:~/Desktop/c/linking$ ./hello-static
Hello, World!
os2024@os2024-virtualbox:~/Desktop/c/linking$ 
```

3) LINKING: STATIC LINKING

- When you open the executable file (which is bigger because it includes the library functions) with a text editor:

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ vi hello-static
```

- The resulting file is not ”human readable” **HOWEVER** there are some symbols that are readable and give interesting information on the executable format, content and needed libraries


"hello-static" [noeol][converted] 2495L, 1035855B
1,1
Top

```
^@f^O^_D^@^@ó^O^^úÿ%Î<8f>
^@f^O^_D^@^@ó^O^^úÿ%Æ<8f>
^@f^O^_D^@^@ó^O^^úÿ%¾<8f>
^@f^O^_D^@^@ó^O^^úÿ%¶<8f>
^@f^O^_D^@^@ó^O^^úÿ%®<8f>
^@f^O^_D^@^@ó^O^^úÿ%¦<8f>
^@f^O^_D^@^@ó^O^^úÿ%<9e><8f>
^@f^O^_D^@^@ó^O^^úÿ%<96><8f>
^@f^O^_D^@^@ó^O^^úÿ%<8e><8f>
^@f^O^_D^@^@ó^O^^úÿ%<86><8f>
^@f^O^_D^@^@ó^O^^úÿ%~<8f>
^@f^O^_D^@^@ó^O^^úÿ%v<8f>
^@f^O^_D^@^@ó^O^^úÿ%n<8f>
^@f^O^_D^@^@ó^O^^úÿ%f<8f>
^@f^O^_D^@^@ó^O^^úÿ%^<8f>
^@f^O^_D^@^@ó^O^^úÿ%V<8f>
^@f^O^_D^@^@ó^O^^úÿ%N<8f>
^@f^O^_D^@^@ó^O^^úÿ%F<8f>
^@f^O^_D^@^@ó^O^^úÿ%><8f>
^@f^O^_D^@^@ó^O^^úÿ%6<8f>
^@f^O^_D^@^@ó^O^^úÿ%.<8f>
^@f^O^_D^@^@A÷E^@^@<80>^@^@u!I<8b>½<88>^@^@^@<80>=À<9e>
^@^@<8b>G^Dt^V<85>Àu*1É1öH<89>O^H<89>7H<89>ßè}<9a>^G^@<85>Àu^T1ÒH<89>W^H<87>^G<83>è^A~çè'¥^@^@ëà<83>è^A<89>G^DëØèf^@^@^@A÷^D$^@<80>^@^@u"
I<8b>¼$<88>^@^@^@<80>=g<9e>
^@^@<8b>G^Dt^V<85>Àu*1É1öH<89>O^H<89>7H<89>ßè$<9a>^G^@<85>Àu^T1ÒH<89>W^H<87>^G<83>è^A~çèÎ¤^@^@ëà<83>è^A<89>G^DëØó^O^^úUH<89>åè^E^@^@^@è^@
^@^@^@ó^O^^úUH<89>åATSH<8d>^] û
^@H<81>ì ^@^@^@dH<8b>^D%(^@^@^@H<89>Eè1ÀdL<8b>$%^P^@^@^@L9%<82>û
^@t^^º^A^@^@^@ð^O±^Ukû
^@t^HH<89>ßè¡£^@^@L<89>%bû
^@ÿ^EXû
^@<83>=]û
^@^@u0H<8d>µPÿÿÿAº^H^@^@^@1ÒÇ^EBû
^@^A^@^@^@HÇ<85>Pÿÿÿ ^@^@^@¿^A^@^@^@¸^N^@^@^@^O^E<8b>^E%û
^@<83>ø^Auw<8b>^E^Nû
^@1É<89>^M^Rû
^@ÿÈ<89>^Eþú
^@u^[1ÒH<89>^U÷ú
^@<87>^Eéú
^@ÿÈ~^HH<89>ßèÝ£^@^@¿^F^@^@^@è#^H^F^@dL<8b>$%^P^@^@^@L9%Ëú
                                                               9,1           0%
```

```
^@^@^@binding file %s [%lu] to %s [%lu]: %s symbol `%s'^@^@^@^@^@^@^@check_match^@^@^@^@^@cannot allocate memory in static TLS block^@^@^
@^@^@^@cannot make segment writable for relocation^@^@^@^@^@cannot restore segment prot after reloc^@%s: Symbol `%s' has different size i
n shared object, consider re-linking
^@^@^@^@^@^@^@%s: Symbol `%s' causes overflow in R_X86_64_32 relocation
^@^@^@^@^@^@%s: Symbol `%s' causes overflow in R_X86_64_PC32 relocation
^@^@^@^@%s: IFUNC symbol '%s' referenced in '%s' is defined in the executable and creates an unsatisfiable circular dependency.
^@^@^@^@^@^@^@^@%s: Relink `%s' with `%s' for IFUNC symbol `%s'
^@^@^@^@^@^@^@^@../sysdeps/x86_64/dl-machine.h^@^@ELFW(R_TYPE) (reloc->r_info) == R_X86_64_RELATIVE^@^@^@^@^@^@^@%s: out of memory to sto
re relocation results for %s
^@^@^@headmap.len == archive_stat.st_size^@^@^@^@^@%a%N%f%N%d%N%b%N%s %h %e %r%N%C-%z %T%N%c%N^@^@^@^@^@ISO/IEC JTC1/SC22/WG20 - internat
ionalization^@^@^@C/o Keld Simonsen, Skt. Jorgens Alle 8, DK-1615 Kobenhavn V^@^@^@^@^@messages^@^@^@^@^@^@^@^@buf->write_ptr != buf->wri
te_end^@^@^@^@^@^@^@^@buf->write_ptr < buf->write_end^@buf->base.write_end == buf->fp->_IO_write_end^@^@^@cy == 1 || (p.frac[p.fracsize -
 2] == 0 && p.frac[0] == 0)^@^@^@^@^@^@cy == 0 || p.tmp[p.tmpsize - 1] < 20^@^@^@^@0 <= p.exponent && p.exponent < 3 && p.exponent + to_s
hift < BITS_PER_MP_LIMB^@^@^@p.expsign == 0 || intdig_max == 1^@^@^@^@^@^@^@../stdio-common/printf_fphex.c^@^@Resource temporarily unavai
lable^@^@^@^@^@^@^@^@Inappropriate ioctl for device^@^@Numerical argument out of domain^@^@^@^@^@^@^@^@Too many levels of symbolic links^
@^@^@^@^@^@Value too large for defined data type^@^@^@Can not access a needed shared library^@^@Accessing a corrupted shared library^@^
@^@^@.lib section in a.out corrupted^@Attempting to link in too many shared libraries^@Cannot exec a shared library directly^@^@^@Invalid
 or incomplete multibyte or wide character^@^@^@^@^@^@^@Interrupted system call should be restarted^@^@^@^@^@Socket operation on non-sock
et^@^@Protocol wrong type for socket^@^@Address family not supported by protocol^@^@^@^@^@^@^@^@Cannot assign requested address^@Network
dropped connection on reset^@^@^@^@^@Software caused connection abort^@^@^@^@^@^@^@^@Transport endpoint is already connected^@Transport e
ndpoint is not connected^@^@^@^@^@Cannot send after transport endpoint shutdown^@^@^@Too many references: cannot splice^@^@^@^@^@^@Operat
ion not possible due to RF-kill^@^@^@Memory page has hardware error^@^@../sysdeps/unix/sysv/linux/x86/../sysconf.c^@^@^@^@^@../sysdeps/un
ix/sysv/linux/sysconf-sigstksz.h^@^@^@../sysdeps/unix/sysv/linux/sysconf-pthread_stack_min.h^@^@linux_sysconf^@^@^@(*lp)->l_idx >= 0 && (
*lp)->l_idx < nloaded^@^@^@^@^@jmap->l_idx >= 0 && jmap->l_idx < nloaded^@^@^@^@^@^@^@imap->l_type == lt_loaded && !imap->l_nodelete_acti
ve^@^@^@
file=%s [%lu];  destroying link map
^@^@^@TLS generation counter wrapped!  Please report as described in <https://bugs.launchpad.net/ubuntu/+source/glibc/+bugs>.
^@^@^@^@^@^@^@^@
closing file=%s; direct_opencount=%u
^@^@Fatal glibc error: cannot allocate memory for find-object data
^@new_nlist < ns->_ns_global_scope_alloc^@^@
add %s [%lu] to global scope
^@^@added <= ns->_ns_global_scope_pending_adds^@^@^@^@^@^@opening file=%s [%lu]; direct_opencount=%u

^@^@^@^@no more namespaces available for dlmopen()^@^@^@^@^@^@invalid target namespace in dlmopen()^@^@^@CPU ISA level is lower than requ
ired^@^@^@^@activating NODELETE for %s [%lu]
^@^@^@^@^@^@^@cannot allocate address lookup data^@^@^@^@^@TLS generation counter wrapped!  Please report this.^@^@^@^@update_scopes^@^@^
@_dl_open^@^@^@^@^@^@^@^@ELFW(R_TYPE)(reloc->r_info) == ELF_MACHINE_JMP_SLOT^@^@^@^@^@_dl_fixup^@^@^@^@^@^@^@unsupported version %s of Ve
rneed record^@^@^@^@^@^@^@^@checking for version `%s' in file %s [%lu] required by file %s [%lu]
                                                                                                   1436,1        57%
```

```
w^@^@^Q^B^R^@<80>kJ^@^@^@^@^@¸^@^@^@^@^@^@^@"w^@^@^Q^B^U^@`±J^@^@^@^@^@ ^@^@^@^@^@^@^@)w^@^@^R^B^G^@<90>ìE^@^@^@^@^@¢^F^@^@^@^@^@^@Fw^@^@
^R^B^G^@°:G^@^@^@^@^@è^@^@^@^@^@^@^@<9f>0^@^@"^@^G^@àÚC^@^@^@^@^@ö^@^@^@^@^@^@^@^@crt1.o^@__abi_tag^@ioputs.o^@_IO_puts.cold^@libc_fatal.
o^@__libc_message_impl.cold^@__PRETTY_FUNCTION__.0^@_IO_new_file_underflow.cold^@loadmsgcat.o^@lock.0^@_nl_load_domain.cold^@abort.o^@sta
ge^@iofclose.o^@_IO_new_fclose.cold^@iofputs.o^@_IO_fputs.cold^@iofwrite.o^@_IO_fwrite.cold^@iogetdelim.o^@__getdelim.cold^@wfileops.o^@a
djust_wide_data^@_IO_wfile_underflow.cold^@pthread_once.o^@__pthread_once_slow^@clear_once_control^@__pthread_cleanup_combined_routine_vo
idptr^@__pthread_once_slow.cold^@printf_fp.o^@hack_digit^@__printf_fp_buffer_1.isra.0^@__printf_fp_buffer_1.isra.0.cold^@printf_fphex.o^@
__printf_fphex_buffer^@__printf_fphex_buffer.cold^@nptl-stack.o^@__nptl_free_stacks.cold^@__printf_buffer_flush.cold^@wprintf_buffer_flus
h.o^@__wprintf_buffer_flush.cold^@unwind-dw2.o^@init_dwarf_reg_size_table^@uw_install_context_1^@uw_install_context_1.cold^@read_encoded_
value^@read_encoded_value.cold^@execute_stack_op^@execute_stack_op.cold^@uw_update_context_1^@uw_update_context_1.cold^@execute_cfa_progr
am_specialized^@execute_cfa_program_specialized.cold^@execute_cfa_program_generic^@execute_cfa_program_generic.cold^@uw_frame_state_for^@
uw_frame_state_for.cold^@uw_init_context_1^@once_regsizes.0^@uw_init_context_1.cold^@_Unwind_RaiseException_Phase2^@_Unwind_RaiseExceptio
n_Phase2.cold^@_Unwind_ForcedUnwind_Phase2^@_Unwind_ForcedUnwind_Phase2.cold^@_Unwind_GetGR.cold^@_Unwind_SetGR.cold^@_Unwind_DebugHook^@
_Unwind_RaiseException.cold^@_Unwind_Resume.cold^@_Unwind_Resume_or_Rethrow.cold^@_Unwind_Backtrace.cold^@unwind-dw2-fde-dip.o^@fde_unenc
oded_compare^@frame_downheap^@version_lock_lock_exclusive^@version_lock_mutex^@version_lock_cond^@read_encoded_value_with_base^@read_enco
ded_value_with_base.cold^@fde_unencoded_extract^@fde_radixsort^@version_lock_unlock_exclusive^@btree_allocate_node^@btree_handle_root_spl
it.part.0^@btree_insert.isra.0^@btree_release_tree_recursively^@btree_destroy^@get_cie_encoding^@btree_remove^@fde_mixed_encoding_extract
^@fde_mixed_encoding_extract.cold^@classify_object_over_fdes^@classify_object_over_fdes.cold^@get_pc_range^@__deregister_frame_info_bases
.part.0^@registered_objects^@in_shutdown^@__deregister_frame_info_bases.part.0.cold^@fde_single_encoding_extract^@fde_single_encoding_ext
ract.cold^@fde_single_encoding_compare^@fde_single_encoding_compare.cold^@fde_mixed_encoding_compare^@fde_mixed_encoding_compare.cold^@ad
d_fdes.isra.0^@add_fdes.isra.0.cold^@linear_search_fdes^@linear_search_fdes.cold^@release_registered_frames^@object_mutex^@terminator.1^@
_Unwind_Find_FDE.cold^@unwind-c.o^@base_of_encoded_value^@base_of_encoded_value.cold^@__gcc_personality_v0.cold^@crtstuff.c^@__EH_FRAME_B
EGIN__^@deregister_tm_clones^@__do_global_dtors_aux^@completed.1^@__do_global_dtors_aux_fini_array_entry^@frame_dummy^@object.0^@__frame_
dummy_init_array_entry^@hello-world.c^@libc-start.o^@handle_zhaoxin^@handle_amd^@_dl_tunable_set_prefer_map_32bit_exec^@__libc_start_call
_main^@get_common_indices.constprop.0^@get_common_cache_info.constprop.0^@intel_check_word.constprop.0^@intel_02_known^@handle_intel.cons
tprop.0^@update_active.constprop.0^@init_cpu_features.constprop.0^@CSWTCH.86^@libc-tls.o^@static_slotinfo^@cxa_atexit.o^@__PRETTY_FUNCTIO
N__.1^@flush_cleanup^@run_fp^@list_all_lock^@save_for_backup^@_IO_un_link.part.0^@stdio_needs_locking^@freeres_list^@dealloc_buffers^@std
files.o^@_IO_stdfile_2_lock^@_IO_wide_data_2^@_IO_stdfile_1_lock^@_IO_wide_data_1^@_IO_stdfile_0_lock^@_IO_wide_data_0^@strops.o^@enlarge
_userbuf^@malloc.o^@_dl_tunable_set_arena_max^@mp_^@_dl_tunable_set_arena_test^@_dl_tunable_set_tcache_max^@_dl_tunable_set_tcache_count^
@_dl_tunable_set_tcache_unsorted_limit^@malloc_printerr^@int_mallinfo^@main_arena^@_dl_tunable_set_mxfast^@global_max_fast^@_dl_tunable_s
et_trim_threshold^@_dl_tunable_set_top_pad^@_dl_tunable_set_mmap_threshold^@_dl_tunable_set_mmaps_max^@_dl_tunable_set_perturb_byte^@allo
c_perturb^@_dl_tunable_set_hugetlb^@detach_arena.part.0^@__PRETTY_FUNCTION__.4^@get_free_list^@thread_arena^@free_list_lock^@__PRETTY_FUN
CTION__.3^@munmap_chunk^@__PRETTY_FUNCTION__.13^@__malloc_info.part.0^@systrim.constprop.0^@__always_fail_morecore^@unlink_chunk.isra.0^@
malloc_consolidate^@_int_free_maybe_consolidate^@aligned_heap_area^@may_shrink_heap.12^@__PRETTY_FUNCTION__.11^@__PRETTY_FUNCTION__.10^@p
tmalloc_init.part.0^@tcache_key^@__malloc_initialized^@_int_free_merge_chunk^@_int_free^@tcache^@alloc_new_heap^@sysmalloc_mmap_fallback.
constprop.0^@arena_get2.part.0^@narenas_limit.2^@narenas^@next_to_use.5^@__PRETTY_FUNCTION__.6^@arena_get_retry^@sysmalloc_mmap.isra.0^@_
_PRETTY_FUNCTION__.9^@_int_malloc^@__PRETTY_FUNCTION__.7^@__PRETTY_FUNCTION__.8^@tcache_init.part.0^@_int_realloc^@__PRETTY_FUNCTION__.16
^@_int_memalign^@__PRETTY_FUNCTION__.18^@tcache_shutting_down^@_mid_memalign.isra.0^@__PRETTY_FUNCTION__.17^@__PRETTY_FUNCTION__.14^@__PR
ETTY_FUNCTION__.15^@__PRETTY_FUNCTION__.19^@__PRETTY_FUNCTION__.20^@memcpy.o^@__new_memcpy_ifunc^@memmove.o^@__libc_memmove_ifunc^@mempcp
y.o^@__mempcpy_ifunc^@memset.o^@memset_ifunc^@strchrnul.o^@__strchrnul_ifunc^@strlen.o^@strlen_ifunc^@strstr.o^@two_way_long_needle^@__li
                                                                                                               2482,1         99%
```

4) CREATING LIBRARIES

- ”my-lib-2024.c”:

```c
#include <stdio.h>

int printHelloNames(char *fname, char *lname) {
        printf("Hello %s %s, nice to meet you!\n", fname, lname);
        return 0;
}

int printJU() {
        printf("This is JU!\n");
        return 0;
}

```

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ gcc -c my-lib-2024.c
os2024@os2024-virtualbox:~/Desktop/c/linking$ ls -la
total 804
drwxrwxr-x 2 os2024 os2024   4096 okt 22 16:12 .
drwxrwxr-x 3 os2024 os2024   4096 okt 22 15:28 ..
-rwxrwxr-x 1 os2024 os2024  15960 okt 22 15:30 hello
-rwxrwxr-x 1 os2024 os2024 785360 okt 22 15:49 hello-static
-rw-rw-r-- 1 os2024 os2024     76 okt 22 15:28 hello-world.c
-rw-rw-r-- 1 os2024 os2024    197 okt 22 16:10 my-lib-2024.c
-rw-rw-r-- 1 os2024 os2024   1768 okt 22 16:12 my-lib-2024.o
os2024@os2024-virtualbox:~/Desktop/c/linking$
```

4) CREATING LIBRARIES

- You build a static library (extension ”.a”) based on your library file using ar:

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ ar rcs libmylib2024.a my-lib-2024.o
```

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ ls -l
total 804
-rwxrwxr-x 1 os2024 os2024  15960 okt 22 15:30 hello
-rwxrwxr-x 1 os2024 os2024 785360 okt 22 15:49 hello-static
-rw-rw-r-- 1 os2024 os2024     76 okt 22 15:28 hello-world.c
-rw-rw-r-- 1 os2024 os2024   1932 okt 22 16:37 libmylib2024.a
-rw-rw-r-- 1 os2024 os2024    197 okt 22 16:10 my-lib-2024.c
-rw-rw-r-- 1 os2024 os2024     84 okt 22 16:15 my-lib-2024.h
-rw-rw-r-- 1 os2024 os2024   1768 okt 22 16:12 my-lib-2024.o
os2024@os2024-virtualbox:~/Desktop/c/linking$
```

4) CREATING LIBRARIES

- ”my-lib-2024.h”:

```c
#include <stdio.h>

int printHelloNames(char *fname, char *lname);

int printJU();

```

- This is the header file containing only the definition of the functions, no code inside!
- Using the static library by creating a ”use-my-lib.c” file:

```c
#include "my-lib-2024.h"

int main() {
        printf("---> Welcome!\n");
        printJU();
        printHelloNames("Jerome", "Landre");
        printf("I hope you are well.   :)\n");
        return 0;
}

```

4) CREATING LIBRARIES

- Using the static library:

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ ls -l
total 808
-rwxrwxr-x 1 os2024 os2024  15960 okt 22 15:30 hello
-rwxrwxr-x 1 os2024 os2024 785360 okt 22 15:49 hello-static
-rw-rw-r-- 1 os2024 os2024     76 okt 22 15:28 hello-world.c
-rw-rw-r-- 1 os2024 os2024   1932 okt 22 16:37 libmylib2024.a
-rw-rw-r-- 1 os2024 os2024    197 okt 22 16:10 my-lib-2024.c
-rw-rw-r-- 1 os2024 os2024     84 okt 22 16:15 my-lib-2024.h
-rw-rw-r-- 1 os2024 os2024   1768 okt 22 16:12 my-lib-2024.o
-rw-rw-r-- 1 os2024 os2024    170 okt 22 16:19 use-my-lib.c
os2024@os2024-virtualbox:~/Desktop/c/linking$ gcc -o use-my-lib use-my-lib.c -I./ -L./ -lmylib2024
os2024@os2024-virtualbox:~/Desktop/c/linking$ ls -l
total 824
-rwxrwxr-x 1 os2024 os2024  15960 okt 22 15:30 hello
-rwxrwxr-x 1 os2024 os2024 785360 okt 22 15:49 hello-static
-rw-rw-r-- 1 os2024 os2024     76 okt 22 15:28 hello-world.c
-rw-rw-r-- 1 os2024 os2024   1932 okt 22 16:37 libmylib2024.a
-rw-rw-r-- 1 os2024 os2024    197 okt 22 16:10 my-lib-2024.c
-rw-rw-r-- 1 os2024 os2024     84 okt 22 16:15 my-lib-2024.h
-rw-rw-r-- 1 os2024 os2024   1768 okt 22 16:12 my-lib-2024.o
-rwxrwxr-x 1 os2024 os2024  16112 okt 22 16:45 use-my-lib
-rw-rw-r-- 1 os2024 os2024    170 okt 22 16:19 use-my-lib.c
os2024@os2024-virtualbox:~/Desktop/c/linking$ ./use-my-lib
---> Welcome!
This is JU!
Hello Jerome Landre, nice to meet you!
I hope you are well.   :)
os2024@os2024-virtualbox:~/Desktop/c/linking$
```

4) CREATING LIBRARIES

- Creating a dynamic (extension ”.so”) library:

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ gcc -shared -o libmylib2024.so my-lib-2024.o
os2024@os2024-virtualbox:~/Desktop/c/linking$ rm use-my-lib2
os2024@os2024-virtualbox:~/Desktop/c/linking$ ls -l
total 840
-rwxrwxr-x 1 os2024 os2024  15960 okt 22 15:30 hello
-rwxrwxr-x 1 os2024 os2024 785360 okt 22 15:49 hello-static
-rw-rw-r-- 1 os2024 os2024     76 okt 22 15:28 hello-world.c
-rw-rw-r-- 1 os2024 os2024   1932 okt 22 16:37 libmylib2024.a
-rwxrwxr-x 1 os2024 os2024  15640 okt 22 16:52 libmylib2024.so
-rw-rw-r-- 1 os2024 os2024    197 okt 22 16:10 my-lib-2024.c
-rw-rw-r-- 1 os2024 os2024     84 okt 22 16:15 my-lib-2024.h
-rw-rw-r-- 1 os2024 os2024   1768 okt 22 16:12 my-lib-2024.o
-rwxrwxr-x 1 os2024 os2024  16112 okt 22 16:45 use-my-lib
-rw-rw-r-- 1 os2024 os2024    170 okt 22 16:19 use-my-lib.c
os2024@os2024-virtualbox:~/Desktop/c/linking$
```

4) CREATING LIBRARIES

- Adding your dynamic library into the system: first, you need to know where are the dynamic libraries:

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ ldconfig -v | less
```

```
/lib/x86_64-linux-gnu: (from /etc/ld.so.conf.d/x86_64-linux-gnu.conf:3)
        libimagequant.so.0 -> libimagequant.so.0
        libargon2.so.1 -> libargon2.so.1
        libsnappy.so.1 -> libsnappy.so.1.1.10
        libnssutil3.so -> libnssutil3.so
        libthread_db.so.1 -> libthread_db.so.1
        libplds4.so -> libplds4.so
        libKF5Plasma.so.5 -> libKF5Plasma.so.5.115.0
```

• • •

```
/lib: (from <builtin>:0)
        libwvdbus.so.4.6 -> libwvdbus.so.4.6
        libwvstreams.so.4.6 -> libwvstreams.so.4.6
        libuniconf.so.4.6 -> libuniconf.so.4.6
        libepub.so.0 -> libepub.so.0.2.1
        libwvutils.so.4.6 -> libwvutils.so.4.6
        libhardsid-builder.so.0 -> libhardsid-builder.so.0.0.1
        libresid-builder.so.0 -> libresid-builder.so.0.0.1
        libchm.so.1 -> libchm.so.1.0.0
        libwvbase.so.4.6 -> libwvbase.so.4.6
        libsidplay2.so.1 -> libsidplay2.so.1.0.1
(END)
```

4) CREATING LIBRARIES

- It comes from the ”/etc/ld.so.conf.d” directory:

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ ls -l /etc/ld.so.conf.d/
total 12
-rw-r--r-- 1 root root  38 jan 21  2024 fakeroot-x86_64-linux-gnu.conf
-rw-r--r-- 1 root root  44 aug  2  2022 libc.conf
-rw-r--r-- 1 root root 100 mar 30  2024 x86_64-linux-gnu.conf
os2024@os2024-virtualbox:~/Desktop/c/linking$ more /etc/ld.so.conf.d/libc.conf
# libc default configuration
/usr/local/lib
os2024@os2024-virtualbox:~/Desktop/c/linking$
```

4) CREATING LIBRARIES

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ sudo ldconfig -v | grep libmylib2024
[sudo] password for os2024:
/sbin/ldconfig.real: Can't stat /usr/local/lib/x86_64-linux-gnu: No such file or directory
/sbin/ldconfig.real: Path `/usr/lib/x86_64-linux-gnu' given more than once
(from /etc/ld.so.conf.d/x86_64-linux-gnu.conf:4 and /etc/ld.so.conf.d/x86_64-linux-gnu.conf:3)
/sbin/ldconfig.real: Path `/lib/x86_64-linux-gnu' given more than once
(from <builtin>:0 and /etc/ld.so.conf.d/x86_64-linux-gnu.conf:3)
/sbin/ldconfig.real: Path `/usr/lib/x86_64-linux-gnu' given more than once
(from <builtin>:0 and /etc/ld.so.conf.d/x86_64-linux-gnu.conf:3)
/sbin/ldconfig.real: Path `/usr/lib' given more than once
(from <builtin>:0 and <builtin>:0)
/sbin/ldconfig.real: /lib/x86_64-linux-gnu/ld-linux-x86-64.so.2 is the dynamic linker, ignoring

os2024@os2024-virtualbox:~/Desktop/c/linking$ sudo cp libmylib2024.so /usr/local/lib
os2024@os2024-virtualbox:~/Desktop/c/linking$ sudo ldconfig -v | grep libmylib2024
/sbin/ldconfig.real: Can't stat /usr/local/lib/x86_64-linux-gnu: No such file or directory
/sbin/ldconfig.real: Path `/usr/lib/x86_64-linux-gnu' given more than once
(from /etc/ld.so.conf.d/x86_64-linux-gnu.conf:4 and /etc/ld.so.conf.d/x86_64-linux-gnu.conf:3)
/sbin/ldconfig.real: Path `/lib/x86_64-linux-gnu' given more than once
(from <builtin>:0 and /etc/ld.so.conf.d/x86_64-linux-gnu.conf:3)
/sbin/ldconfig.real: Path `/usr/lib/x86_64-linux-gnu' given more than once
(from <builtin>:0 and /etc/ld.so.conf.d/x86_64-linux-gnu.conf:3)
/sbin/ldconfig.real: Path `/usr/lib' given more than once
(from <builtin>:0 and <builtin>:0)
        libmylib2024.so -> libmylib2024.so
/sbin/ldconfig.real: /lib/x86_64-linux-gnu/ld-linux-x86-64.so.2 is the dynamic linker, ignoring

os2024@os2024-virtualbox:~/Desktop/c/linking$
```

4) CREATING LIBRARIES

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ gcc -o use-my-lib2 use-my-lib.c -l mylib2024
os2024@os2024-virtualbox:~/Desktop/c/linking$ ls -l
total 856
-rwxrwxr-x 1 os2024 os2024  15960 okt 22 15:30 hello
-rwxrwxr-x 1 os2024 os2024 785360 okt 22 15:49 hello-static
-rw-rw-r-- 1 os2024 os2024     76 okt 22 15:28 hello-world.c
-rw-rw-r-- 1 os2024 os2024   1932 okt 22 16:37 libmylib2024.a
-rwxrwxr-x 1 os2024 os2024  15640 okt 22 16:52 libmylib2024.so
-rw-rw-r-- 1 os2024 os2024    197 okt 22 16:10 my-lib-2024.c
-rw-rw-r-- 1 os2024 os2024     84 okt 22 16:15 my-lib-2024.h
-rw-rw-r-- 1 os2024 os2024   1768 okt 22 16:12 my-lib-2024.o
-rwxrwxr-x 1 os2024 os2024  16112 okt 22 16:45 use-my-lib
-rwxrwxr-x 1 os2024 os2024  16032 okt 22 17:06 use-my-lib2
-rw-rw-r-- 1 os2024 os2024    170 okt 22 16:19 use-my-lib.c
os2024@os2024-virtualbox:~/Desktop/c/linking$ ./use-my-lib2
---> Welcome!
This is JU!
Hello Jerome Landre, nice to meet you!
I hope you are well.   :)
os2024@os2024-virtualbox:~/Desktop/c/linking$
```

4) CREATING LIBRARIES

- Just to be sure we are using the library in ”/usr/local/lib”:

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ rm libmylib2024.so
```

```
os2024@os2024-virtualbox:~/Desktop/c/linking$ ls -l
total 840
-rwxrwxr-x 1 os2024 os2024  15960 okt 22 15:30 hello
-rwxrwxr-x 1 os2024 os2024 785360 okt 22 15:49 hello-static
-rw-rw-r-- 1 os2024 os2024     76 okt 22 15:28 hello-world.c
-rw-rw-r-- 1 os2024 os2024   1932 okt 22 16:37 libmylib2024.a
-rw-rw-r-- 1 os2024 os2024    197 okt 22 16:10 my-lib-2024.c
-rw-rw-r-- 1 os2024 os2024     84 okt 22 16:15 my-lib-2024.h
-rw-rw-r-- 1 os2024 os2024   1768 okt 22 16:12 my-lib-2024.o
-rwxrwxr-x 1 os2024 os2024  16112 okt 22 16:45 use-my-lib
-rwxrwxr-x 1 os2024 os2024  16032 okt 22 17:12 use-my-lib2
-rw-rw-r-- 1 os2024 os2024    170 okt 22 16:19 use-my-lib.c
os2024@os2024-virtualbox:~/Desktop/c/linking$ gcc -o use-my-lib2 use-my-lib.c -l mylib2024
os2024@os2024-virtualbox:~/Desktop/c/linking$ ./use-my-lib2
---> Welcome!
This is JU!
Hello Jerome Landre, nice to meet you!
I hope you are well.   :)
os2024@os2024-virtualbox:~/Desktop/c/linking$ 
```

5) MAKE

- ”make” is a command to run commands automatically
- You need to create a Makefile file to use ”make”

```
target: source1 source2 ...
        command1
        command2
        ...
```

- ”make target” to run the associated commands

5) MAKE

- all: make everything:

```
all: client server
```

- clean: remove generated files, start over with just source

```
clean:
	rm -f *.o client server
```

5) MAKE

- ”make” is a command to run commands automatically
- You need to create a Makefile file to use ”make”

```
hello: hello.c
        gcc –g –Wall –o hello hello.c

hello2: hello2.c lib.c lib.h
        gcc –g –Wall –o hello2 hello2.c lib.c
```

5) MAKE

- ”make” is a command to run commands automatically
- You need to create a Makefile file to use ”make”

```
CC = gcc
CFLAGS = -g -Wall
hello2: hello2.c lib.c lib.h
	$(CC) $(CFLAGS) –o hello2 hello2.c lib.c
```

5) MAKE

- Make can help the developers a lot when dealing with complicated applications:

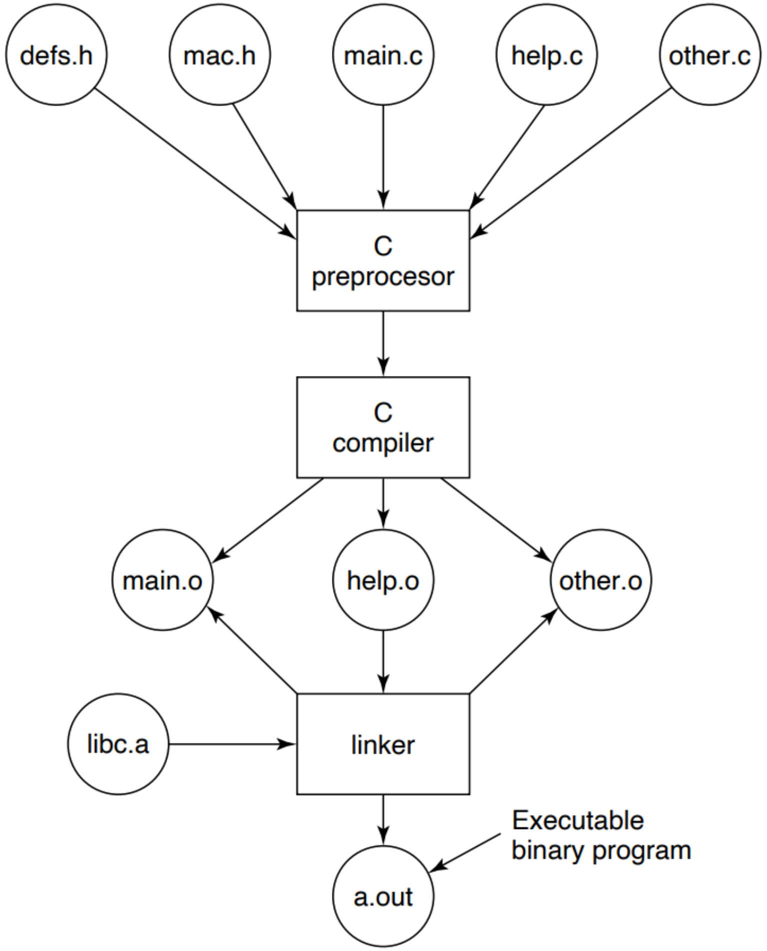


Figure 1-30. The process of compiling C and header files to make an executable.

(Tanenbaum&Bos, ”Modern OperatingSystems, fifth edition”, Pearson, 2024)

ANY QUESTIONS?